

```
In [189... import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

sns.set(style="whitegrid")
```

```
In [190... df = pd.read_csv("WorldEnergy_.csv")

df.head()
```

```
Out[190... country year iso_code population gdp biofuel_cons_change_pct biofuel_cons_change
```

	country	year	iso_code	population	gdp	biofuel_cons_change_pct	biofuel_cons_change
0	ASEAN (Ember)	2000	NaN	NaN	NaN	NaN	
1	ASEAN (Ember)	2001	NaN	NaN	NaN	NaN	
2	ASEAN (Ember)	2002	NaN	NaN	NaN	NaN	
3	ASEAN (Ember)	2003	NaN	NaN	NaN	NaN	
4	ASEAN (Ember)	2004	NaN	NaN	NaN	NaN	

5 rows × 130 columns



```
In [191... data = df[['country', 'year', 'gdp', 'nuclear_share_elec', 'greenhouse_gas_emission
```

```
In [192... data = data.dropna(subset=[
    'country',
    'gdp',
    'nuclear_share_elec',
    'greenhouse_gas_emissions'
])
```

```
In [193... exclude_keywords = [
    'World', 'Asia', 'Africa', 'Europe', 'America',
    'income', 'OECD', 'G20', 'G7', 'EU', 'Ember'
]

data = data[
    ~data['country'].str.contains('|'.join(exclude_keywords), case=False, na=False)
].copy()

print("Countries remaining:", data['country'].unique())
```

Countries remaining: 163

```
In [205... data = data.rename(columns={'greenhouse_gas_emissions': 'ghg'})
```

```
In [206... data = data[
    (data['gdp'] > 0) &
    (data['ghg'] > 0) &
    (data['nuclear_share_elec'] > 0)
].copy()
print("Rows after filtering:", len(data))
```

Rows after filtering: 694

```
In [207... data['log_ghg'] = np.log10(data['ghg'])
data['log_gdp'] = np.log10(data['gdp'])
```

```
In [208... data['gdp_group'] = pd.qcut(
    data['gdp'],
    q=3,
    labels=['Low GDP', 'Middle GDP', 'High GDP']
)
```

```
In [209... data['nuclear_group'] = pd.qcut(
    data['nuclear_share_elec'],
    q=3,
    labels=['Low Nuclear', 'Medium Nuclear', 'High Nuclear']
)
```

```
In [210... print(data.groupby(['gdp_group', 'nuclear_group']).size())
```

gdp_group	nuclear_group	
Low GDP	Low Nuclear	11
	Medium Nuclear	66
	High Nuclear	154
Middle GDP	Low Nuclear	115
	Medium Nuclear	62
	High Nuclear	55
High GDP	Low Nuclear	105
	Medium Nuclear	103
	High Nuclear	23

dtype: int64

C:\Users\Chee Zing\AppData\Local\Temp\ipykernel_19612\1416240683.py:1: FutureWarning: The default of observed=False is deprecated and will be changed to True in a future version of pandas. Pass observed=False to retain current behavior or observed=True to adopt the future default and silence this warning.

```
print(data.groupby(['gdp_group', 'nuclear_group']).size())
```

```
In [211... print(data.info())
```

```

<class 'pandas.core.frame.DataFrame'>
Index: 694 entries, 1126 to 21818
Data columns (total 9 columns):
#   Column                Non-Null Count  Dtype
---  -
0   country                694 non-null   object
1   year                   694 non-null   int64
2   gdp                    694 non-null   float64
3   nuclear_share_elec     694 non-null   float64
4   ghg                    694 non-null   float64
5   log_ghg                694 non-null   float64
6   log_gdp                694 non-null   float64
7   gdp_group              694 non-null   category
8   nuclear_group          694 non-null   category
dtypes: category(2), float64(5), int64(1), object(1)
memory usage: 45.0+ KB
None

```

In [212... `print(data[['gdp', 'nuclear_share_elec', 'ghg']].describe())`

```

          gdp  nuclear_share_elec      ghg
count  6.940000e+02      694.000000  694.000000
mean    2.257191e+12      25.664487  297.344092
std     4.010486e+12      19.705998  715.385790
min     1.524483e+10       0.047000   0.730000
25%     2.836155e+11       6.675750  16.635000
50%     8.235258e+11      20.822500  53.805000
75%     2.296028e+12      37.972500 184.830000
max     2.696602e+13      82.253000 5191.540000

```

In [213... `print("GDP group distribution:")`
`print(data['gdp_group'].value_counts())`

`print("\nNuclear group distribution:")`
`print(data['nuclear_group'].value_counts())`

```

GDP group distribution:
gdp_group
Middle GDP    232
Low GDP       231
High GDP      231
Name: count, dtype: int64

```

```

Nuclear group distribution:
nuclear_group
High Nuclear    232
Low Nuclear     231
Medium Nuclear  231
Name: count, dtype: int64

```

In [214... `print("\nMean GHG by GDP group:")`
`print(data.groupby('gdp_group')['log_ghg'].mean())`

`print("\nMean GHG by Nuclear group:")`
`print(data.groupby('nuclear_group')['log_ghg'].mean())`

```
print("\nMean GHG by both:")
print(data.groupby(['gdp_group', 'nuclear_group'])['log_ghg'].mean())
```

Mean GHG by GDP group:

```
gdp_group
Low GDP      1.009856
Middle GDP   1.738103
High GDP     2.534279
Name: log_ghg, dtype: float64
```

Mean GHG by Nuclear group:

```
nuclear_group
Low Nuclear    2.188681
Medium Nuclear 2.041571
High Nuclear   1.054943
Name: log_ghg, dtype: float64
```

Mean GHG by both:

```
gdp_group  nuclear_group
Low GDP    Low Nuclear    1.411815
           Medium Nuclear  1.190855
           High Nuclear   0.903573
Middle GDP Low Nuclear    1.887074
           Medium Nuclear  1.893305
           High Nuclear   1.251662
High GDP   Low Nuclear    2.600397
           Medium Nuclear  2.675936
           High Nuclear   1.598052
Name: log_ghg, dtype: float64
```

C:\Users\Chee Zing\AppData\Local\Temp\ipykernel_19612\3842249110.py:2: FutureWarning: The default of observed=False is deprecated and will be changed to True in a future version of pandas. Pass observed=False to retain current behavior or observed=True to adopt the future default and silence this warning.

```
print(data.groupby('gdp_group')['log_ghg'].mean())
```

C:\Users\Chee Zing\AppData\Local\Temp\ipykernel_19612\3842249110.py:5: FutureWarning: The default of observed=False is deprecated and will be changed to True in a future version of pandas. Pass observed=False to retain current behavior or observed=True to adopt the future default and silence this warning.

```
print(data.groupby('nuclear_group')['log_ghg'].mean())
```

C:\Users\Chee Zing\AppData\Local\Temp\ipykernel_19612\3842249110.py:8: FutureWarning: The default of observed=False is deprecated and will be changed to True in a future version of pandas. Pass observed=False to retain current behavior or observed=True to adopt the future default and silence this warning.

```
print(data.groupby(['gdp_group', 'nuclear_group'])['log_ghg'].mean())
```

In [215...

```
corr = data[['log_gdp', 'nuclear_share_elec', 'log_ghg']].corr()
print(corr)
```

```
          log_gdp  nuclear_share_elec  log_ghg
log_gdp      1.000000         -0.492595  0.892535
nuclear_share_elec -0.492595          1.000000 -0.570577
log_ghg       0.892535         -0.570577  1.000000
```

In [216...

```
print(data.isnull().sum())
```

```
country      0
year         0
gdp          0
nuclear_share_elec  0
ghg          0
log_ghg      0
log_gdp      0
gdp_group    0
nuclear_group 0
dtype: int64
```

In [217... `print(data.isnull().sum())`

```
country      0
year         0
gdp          0
nuclear_share_elec  0
ghg          0
log_ghg      0
log_gdp      0
gdp_group    0
nuclear_group 0
dtype: int64
```

In [218... `Q1 = data['log_ghg'].quantile(0.25)`
`Q3 = data['log_ghg'].quantile(0.75)`
`IQR = Q3 - Q1`

`lower = Q1 - 1.5 * IQR`
`upper = Q3 + 1.5 * IQR`

`outliers = data[`
 `(data['log_ghg'] < lower) |`
 `(data['log_ghg'] > upper)`
`]`

`print("Outliers count:", len(outliers))`

Outliers count: 0

In [219... `mean_gdp_actual = data.groupby('gdp_group')['gdp'].mean()`
`print(mean_gdp_actual)`

```
gdp_group
Low GDP      1.863169e+11
Middle GDP   9.250289e+11
High GDP     5.665994e+12
Name: gdp, dtype: float64
```

C:\Users\Chee Zing\AppData\Local\Temp\ipykernel_19612\3391637652.py:1: FutureWarning: The default of observed=False is deprecated and will be changed to True in a future version of pandas. Pass observed=False to retain current behavior or observed=True to adopt the future default and silence this warning.
`mean_gdp_actual = data.groupby('gdp_group')['gdp'].mean()`

In [220... `mean_nuclear_actual = data.groupby('nuclear_group')['nuclear_share_elec'].mean()`
`print(mean_nuclear_actual)`

```
nuclear_group
Low Nuclear      5.233346
Medium Nuclear   23.268463
High Nuclear     48.393259
Name: nuclear_share_elec, dtype: float64
```

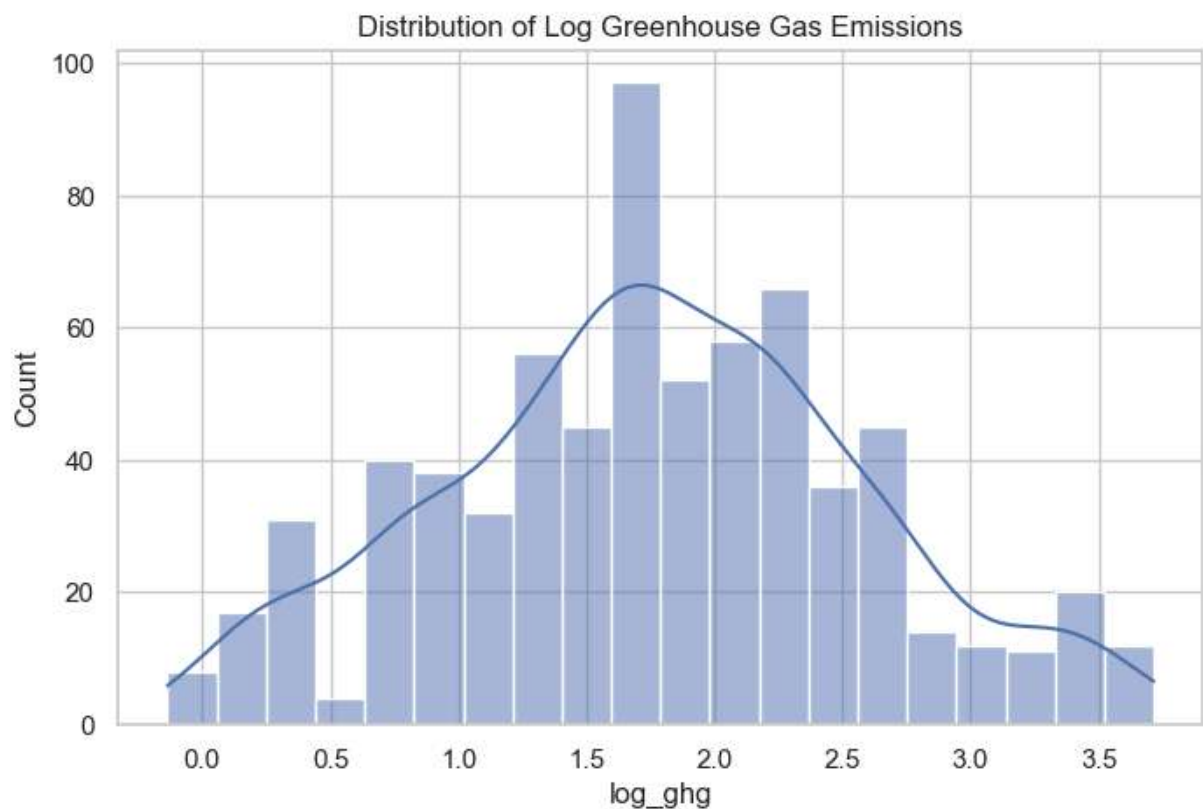
C:\Users\Chee Zing\AppData\Local\Temp\ipykernel_19612\2870614767.py:1: FutureWarning: The default of observed=False is deprecated and will be changed to True in a future version of pandas. Pass observed=False to retain current behavior or observed=True to adopt the future default and silence this warning.

```
mean_nuclear_actual = data.groupby('nuclear_group')['nuclear_share_elec'].mean()
```

In [221...

```
plt.figure(figsize=(8,5))
sns.histplot(data['log_ghg'], bins=20, kde=True)

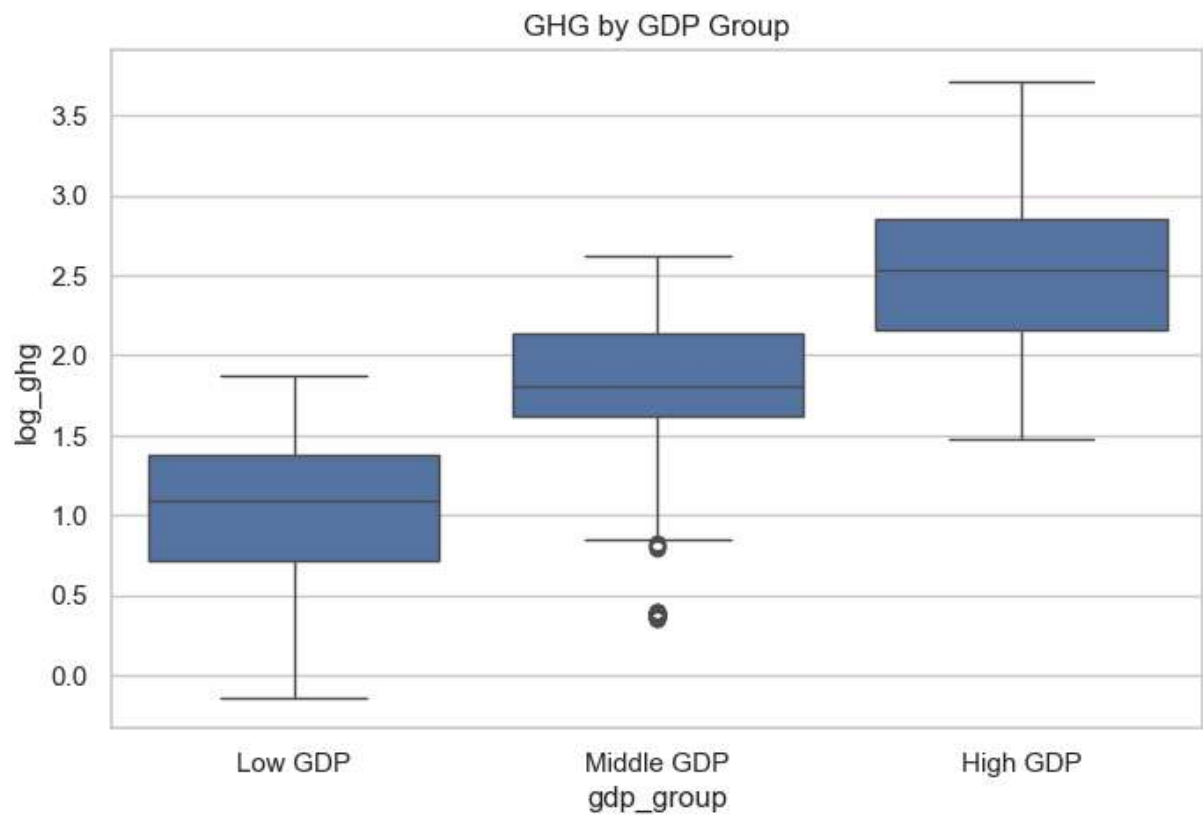
plt.title("Distribution of Log Greenhouse Gas Emissions")
plt.show()
```



In [222...

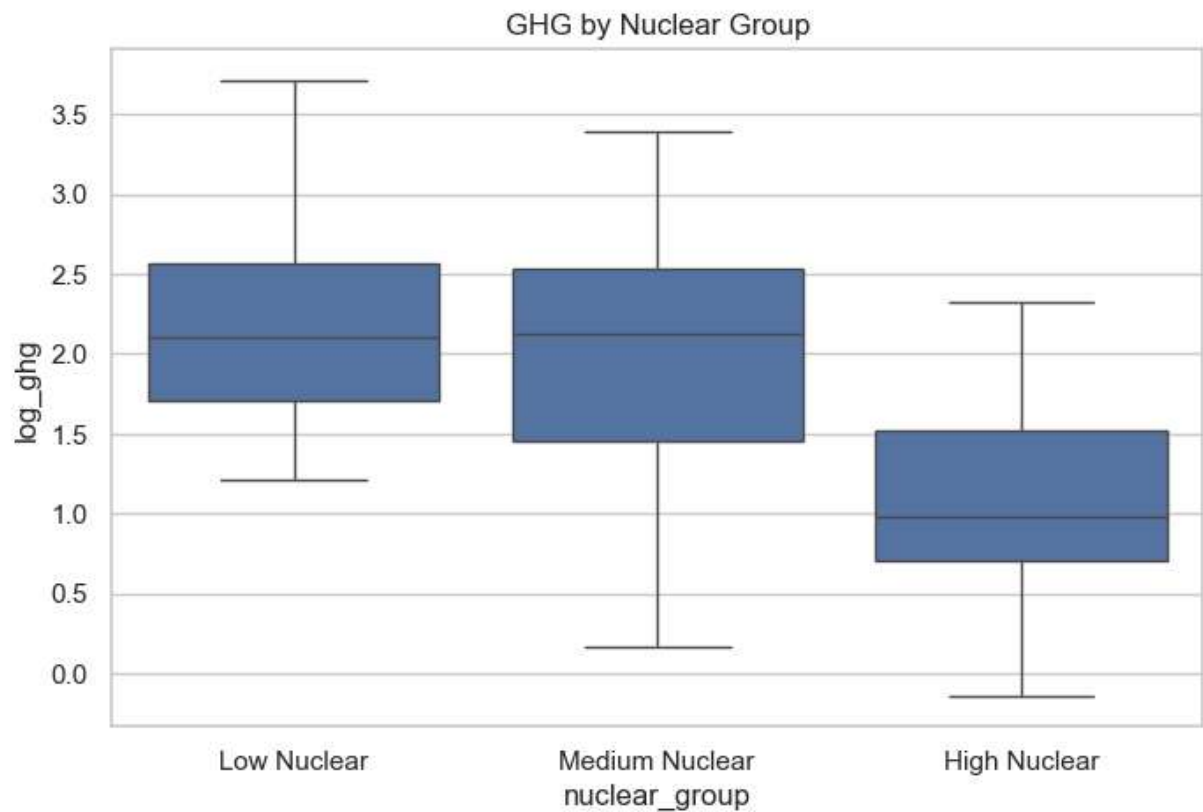
```
plt.figure(figsize=(8,5))
sns.boxplot(x='gdp_group', y='log_ghg', data=data)

plt.title("GHG by GDP Group")
plt.show()
```



```
In [223... plt.figure(figsize=(8,5))
sns.boxplot(x='nuclear_group', y='log_ghg', data=data)

plt.title("GHG by Nuclear Group")
plt.show()
```

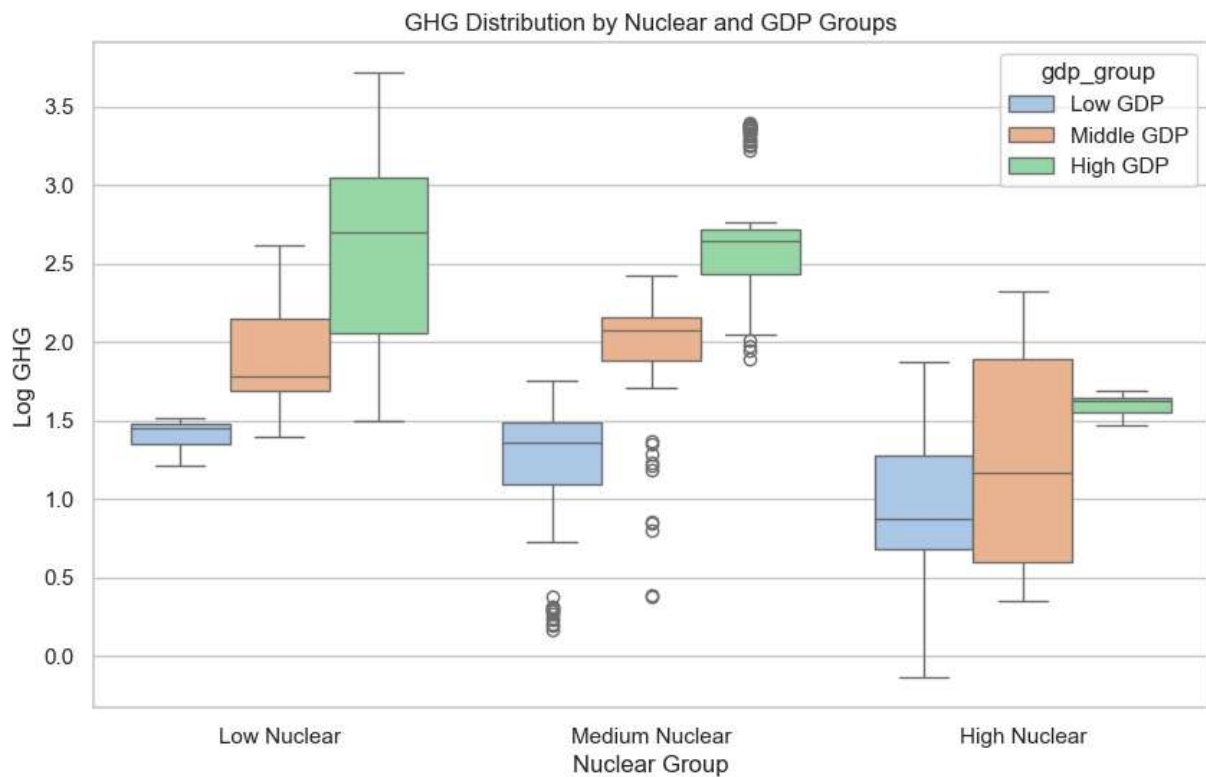


In [224...

```
palette = ['#4C72B0', '#DD8452', '#55A868']
plt.figure(figsize=(10,6))

sns.boxplot(
    x='nuclear_group',
    y='log_ghg',
    hue='gdp_group',
    data=data,
    palette='pastel'
)

plt.title("GHG Distribution by Nuclear and GDP Groups")
plt.xlabel("Nuclear Group")
plt.ylabel("Log GHG")
plt.show()
```



In [225...

```
mean_gdp = data.groupby('gdp_group')['log_ghg'].mean().reset_index()

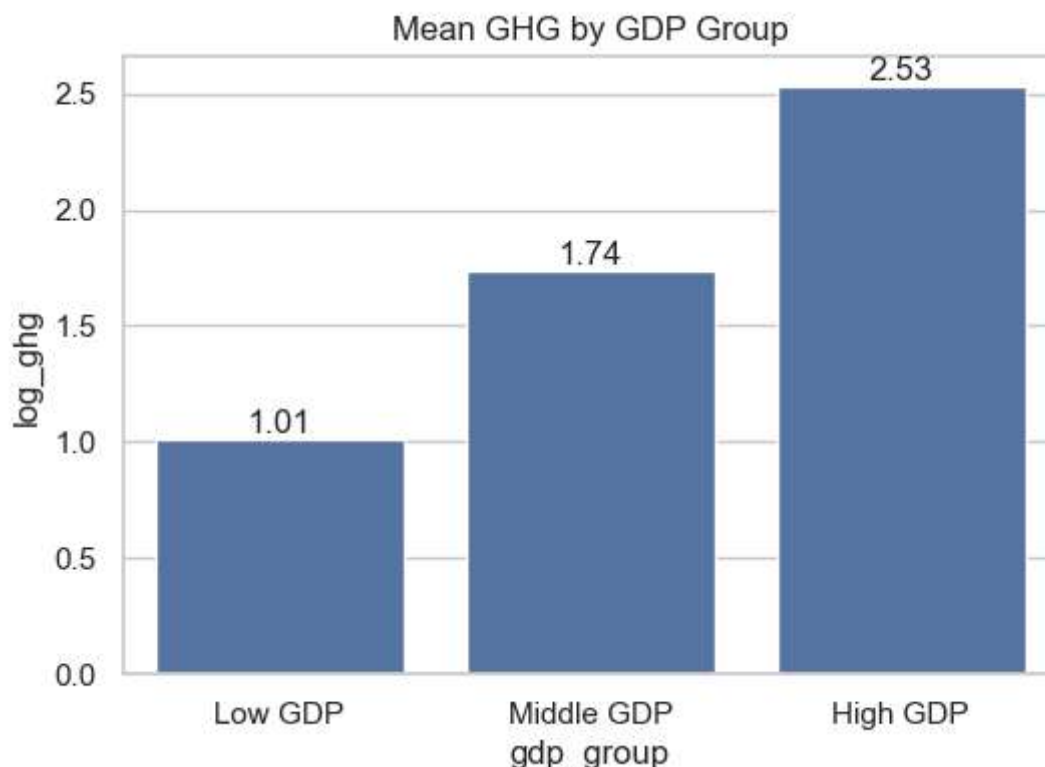
plt.figure(figsize=(6,4))
ax = sns.barplot(x='gdp_group', y='log_ghg', data=mean_gdp)

for c in ax.containers:
    ax.bar_label(c, fmt='%.2f')

plt.title("Mean GHG by GDP Group")
plt.show()
```


C:\Users\Chee Zing\AppData\Local\Temp\ipykernel_19612\2437010207.py:1: FutureWarning: The default of observed=False is deprecated and will be changed to True in a future version of pandas. Pass observed=False to retain current behavior or observed=True to adopt the future default and silence this warning.

```
mean_gdp = data.groupby('gdp_group')['log_ghg'].mean().reset_index()
```



```
In [226... mean_nuclear = data.groupby('nuclear_group')['log_ghg'].mean().reset_index()

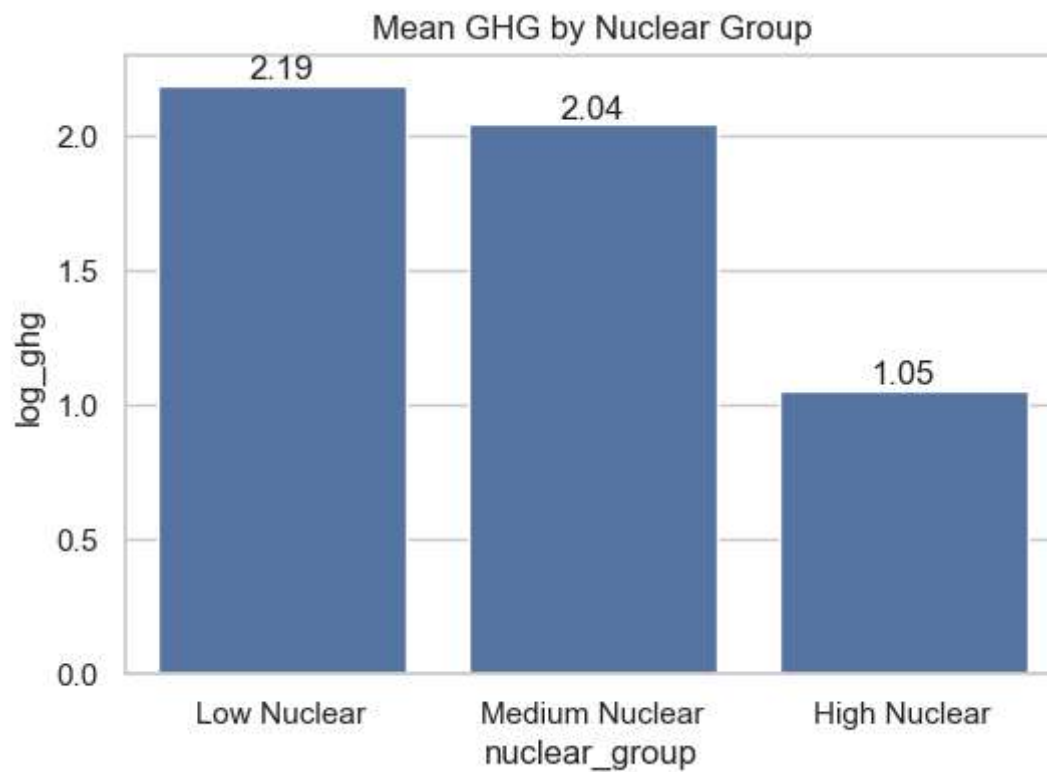
plt.figure(figsize=(6,4))
ax = sns.barplot(x='nuclear_group', y='log_ghg', data=mean_nuclear)

for c in ax.containers:
    ax.bar_label(c, fmt='%.2f')

plt.title("Mean GHG by Nuclear Group")
plt.show()
```

C:\Users\Chee Zing\AppData\Local\Temp\ipykernel_19612\568478793.py:1: FutureWarning: The default of observed=False is deprecated and will be changed to True in a future version of pandas. Pass observed=False to retain current behavior or observed=True to adopt the future default and silence this warning.

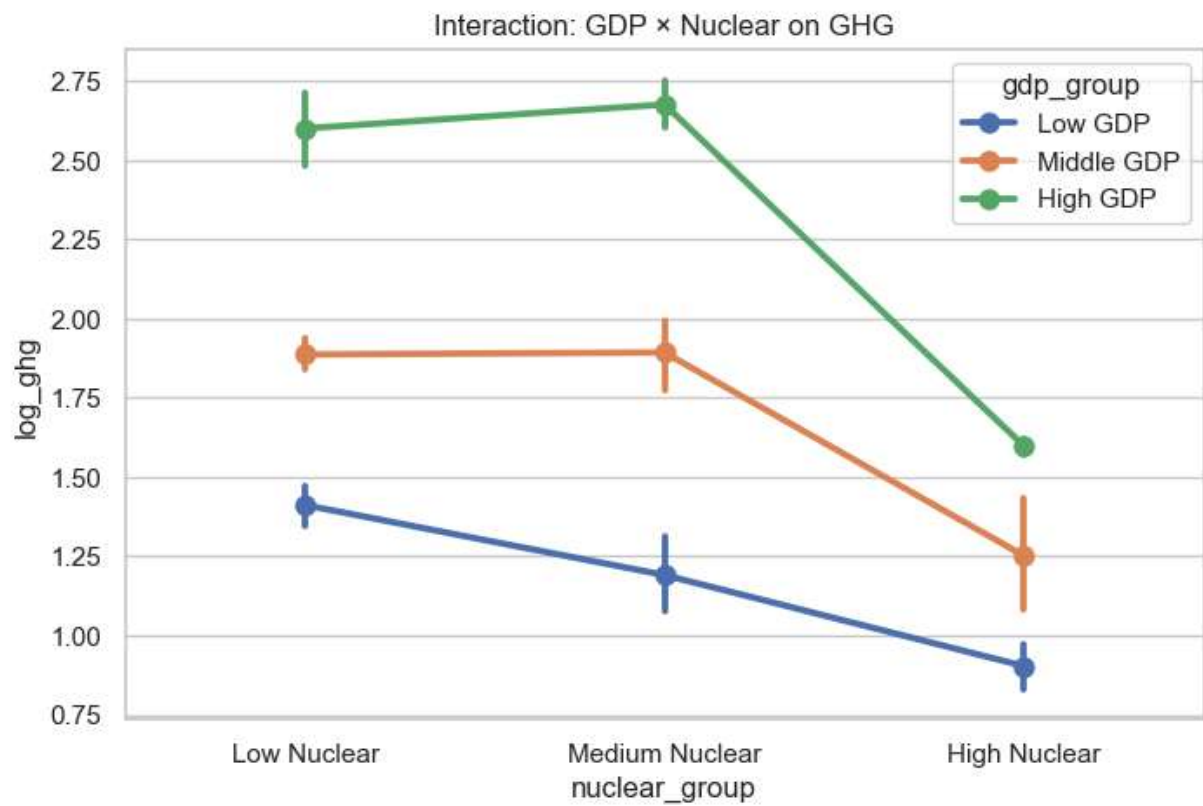
```
mean_nuclear = data.groupby('nuclear_group')['log_ghg'].mean().reset_index()
```



```
In [227... plt.figure(figsize=(8,5))

sns.pointplot(
    x='nuclear_group',
    y='log_ghg',
    hue='gdp_group',
    data=data
)

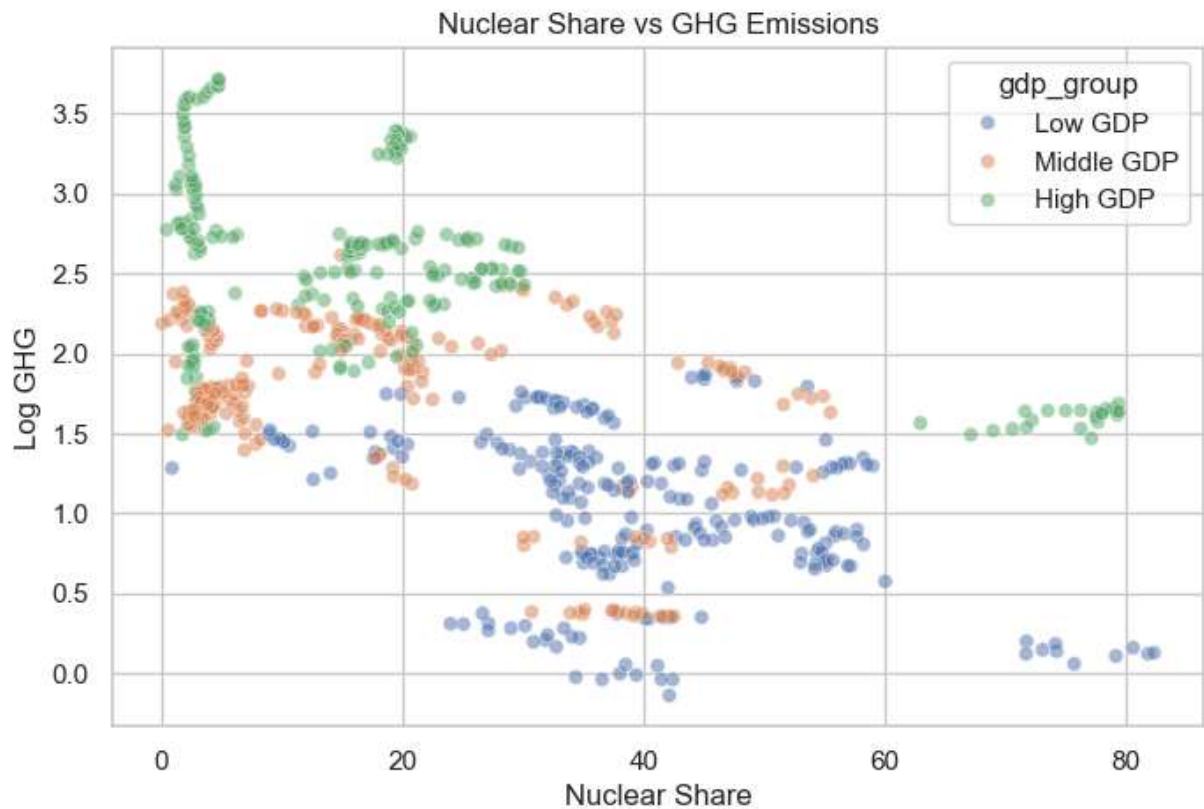
plt.title("Interaction: GDP × Nuclear on GHG")
plt.show()
```



```
In [228... plt.figure(figsize=(8,5))

sns.scatterplot(
    x='nuclear_share_elec',
    y='log_ghg',
    hue='gdp_group',
    data=data,
    alpha=0.5
)

plt.title("Nuclear Share vs GHG Emissions")
plt.xlabel("Nuclear Share")
plt.ylabel("Log GHG")
plt.show()
```



In [229...

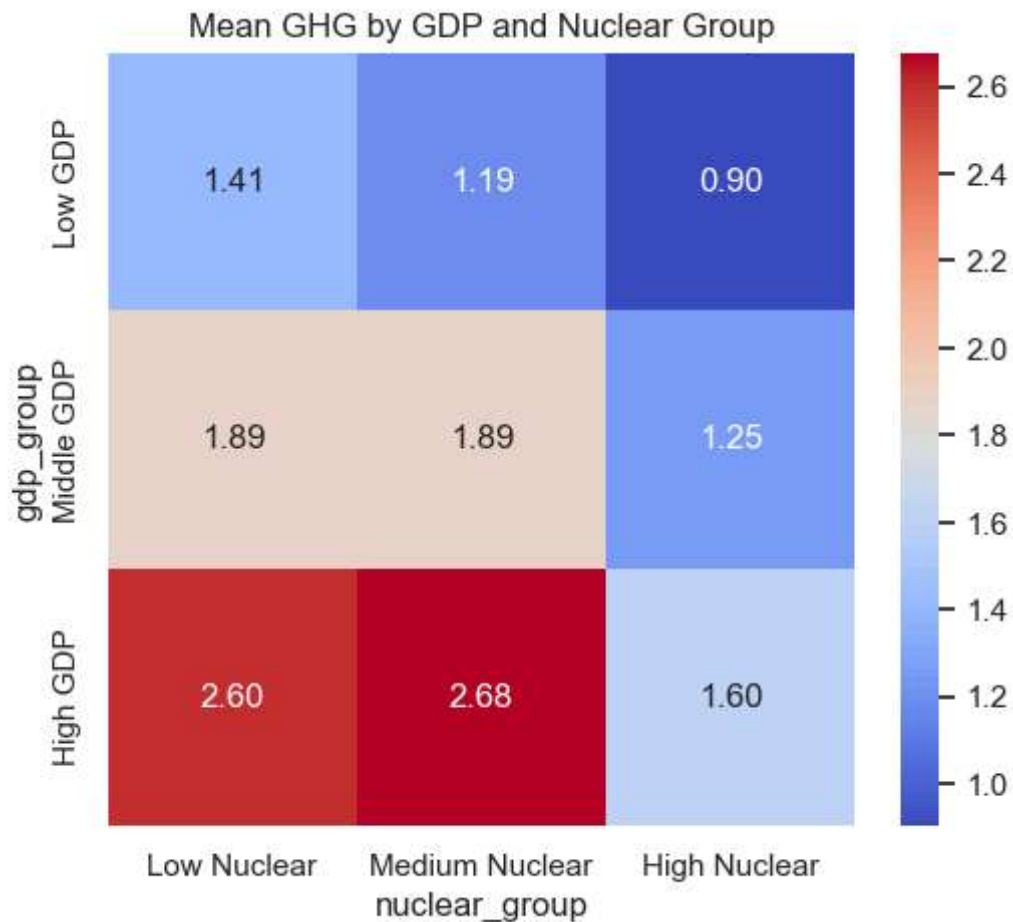
```
pivot = data.pivot_table(
    values='log_ghg',
    index='gdp_group',
    columns='nuclear_group',
    aggfunc='mean'
)

plt.figure(figsize=(6,5))
sns.heatmap(pivot, annot=True, fmt=".2f", cmap="coolwarm")

plt.title("Mean GHG by GDP and Nuclear Group")
plt.show()
```

C:\Users\Chee Zing\AppData\Local\Temp\ipykernel_19612\982426351.py:1: FutureWarning: The default value of observed=False is deprecated and will change to observed=True in a future version of pandas. Specify observed=False to silence this warning and retain the current behavior

```
    pivot = data.pivot_table(
```



```
In [230... from statsmodels.formula.api import ols
import statsmodels.api as sm

model_2way = ols(
    'log_ghg ~ C(gdp_group) * C(nuclear_group)',
    data=data
).fit()

anova_2way = sm.stats.anova_lm(model_2way, typ=2)
print(anova_2way)
```

	sum_sq	df	F	PR(>F)
C(gdp_group)	129.240295	2.0	286.285511	4.306577e-91
C(nuclear_group)	36.747967	2.0	81.401939	1.917992e-32
C(gdp_group):C(nuclear_group)	8.676921	4.0	9.610303	1.451091e-07
Residual	154.617678	685.0	NaN	NaN

```
In [231... from scipy import stats

residuals = model_2way.resid
shapiro_test = stats.shapiro(residuals)

print("Shapiro-Wilk Test:", shapiro_test)
```

Shapiro-Wilk Test: ShapiroResult(statistic=0.9864891971859391, pvalue=5.0435220143686835e-06)

```
In [232... groups = [group["log_ghg"].values for name, group in data.groupby(['gdp_group', 'nuc  
levene_test = stats.levene(*groups)  
  
print("Levene Test:", levene_test)
```

Levene Test: LeveneResult(statistic=14.653871702441133, pvalue=7.103162673256211e-20)

C:\Users\Chee Zing\AppData\Local\Temp\ipykernel_19612\3244460042.py:1: FutureWarning: The default of observed=False is deprecated and will be changed to True in a future version of pandas. Pass observed=False to retain current behavior or observed=True to adopt the future default and silence this warning.

```
groups = [group["log_ghg"].values for name, group in data.groupby(['gdp_group', 'nuclear_group'])]
```

```
In [233... eta_sq = anova_2way['sum_sq'] / sum(anova_2way['sum_sq'])  
print(eta_sq)
```

```
C(gdp_group)                0.392490  
C(nuclear_group)            0.111600  
C(gdp_group):C(nuclear_group) 0.026351  
Residual                    0.469559  
Name: sum_sq, dtype: float64
```